

Linear wave Theory

- Real water waves propagates in a viscous fluid over an irregular bottom of varying permeability.
- However the main body of the fluid is irrotational.

➤ WHY ?

- Because viscous effects are usually concentrate in thin boundary layers near the surface and bottom.
- Since water can also be considered reasonably incompressible, a velocity potential & stream function should exist for wave.

BOUNDARY VALUE PROBLEMS (BVP)

➤ The formulation of BVP is simply the expression of a physical solution in mathematical form such that a unique solution exists.

➤ Steps

- Establishing a region of interest.
- Specifying a differential equation that must be satisfied within the region.

- **Selecting one or more solutions out of infinite number of solutions relevant to physical problem under investigation. This is done using the Boundary Conditions(BC), i.e rejecting those solutions that are incompatible with these conditions.**

➤ **In addition to spatial (geometric) BC, temporal (initial condition) BC should be specified.**

Under the assumption of irrotational motion and incompressible fluid a velocity potential exist which satisfies continuity equation

$$\nabla \cdot \mathbf{U} = 0 \quad (1.a)$$

OR $\nabla \cdot \nabla \phi = 0 \quad (1.b)$

Giving

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \quad (2)$$

➤ For flows that are non-divergent and irrotational Laplace equation also applies to stream function.

$$\frac{\partial w}{\partial x} - \frac{\partial u}{\partial z} = 0 \quad (3.a)$$

Since $u = -\frac{\partial \psi}{\partial z}$ and $w = \frac{\partial \psi}{\partial x}$ (In 2D Plane)

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial z^2} = 0 \quad (3.b)$$

Above equation must hold throughout the fluid.

➤ Question what if the flow was frictionless but rotational what could be the equation (3.b) look like?

➤ Answer

$$\boxed{\nabla^2 \varphi = \omega} \quad \longrightarrow$$

Vorticity

(4)

- 
- Velocity potential can be defined for 2D and 3D whereas the definition of the stream function is such that it only can be defined for 2D and 3D if and only if the flow is symmetric about an axis i.e. is mathematically 2D.
 - Laplace equation is linear i.e. involves no products and thus has a valuable property of superposition.
 - If ϕ_1 and ϕ_2 each satisfy Laplace equation then $\phi_3 = A\phi_1 + B\phi_2$ also satisfies Laplace equation.

KINEMATIC BOUNDARY CONDITIONS(KBC)

- **At any boundary (whether fixed or free) certain physical conditions must be satisfied by fluid velocities. These conditions on water particle kinematics are called kinematic boundary conditions.**
- **At any surface or fluid interface there must be no flow across the interface :
otherwise there would be no interface.**
 - quite obvious in case of an impermeable fixed surface like sea wall.
- **Mathematical expression for KBC may be derived from the equation which describes the surface that constitutes the boundary.**

HOW?

- Any moving or fixed surface is written as

$$F(x, y, z, t) = 0$$

Example: sphere?? (of radius a)

$$F(x, y, z, t) = x^2 + y^2 + z^2 - a^2 = 0$$

For such surface to represent an interface the total derivative or material derivative of the surface would be zero on surface.

$$\left. \frac{DF}{Dt} = \frac{\partial F}{\partial t} + u \frac{\partial F}{\partial x} + v \frac{\partial F}{\partial y} + w \frac{\partial F}{\partial z} \right| = 0 \quad (5.a)$$

on $F=0$ (surface)

Or

$$\frac{-\partial F}{\partial t} = \bar{U} \cdot \nabla F \quad (5.b)$$

If we define unit vector normal to the surface such as $n = \frac{\nabla F}{|\nabla F|}$ Eq. (5.b) can be written as

$$\frac{-\partial F}{\partial t} = \bar{U} \cdot n |\nabla F| \quad (5.c)$$

Or

$$\bar{U} \cdot n = \frac{\frac{-\partial F}{\partial t}}{|\nabla F|} \quad \text{on } F(x, y, z, t) = 0 \quad (6)$$

here

$$|\nabla F| = \sqrt{\left(\frac{\partial F}{\partial x}\right)^2 + \left(\frac{\partial F}{\partial y}\right)^2 + \left(\frac{\partial F}{\partial z}\right)^2}$$

- This condition requires that $(\mathbf{u} \cdot \mathbf{n})$ fluid velocity normal to surface be related to local velocity of surface $\left(\frac{\partial F}{\partial t}\right)$
- If surface do not move with respect to time i.e $\mathbf{u} \cdot \mathbf{n} = 0$ or velocity component normal to surface is zero.

Bottom Boundary Condition (BBC)

- Bottom described as $z=-h(x)$ for 2D case where origin is located at still water level(SWL) & h represents depth.
- Since bottom is fixed $U.n=0$
- Here $F(x,z) = z+h(x)=0$ (7)

&

$$u.n = 0$$

$$U = u\hat{i} + w\hat{k}$$

$$\nabla F = \frac{\partial F}{\partial x} \hat{i} + \frac{\partial F}{\partial z} \hat{k} = \frac{\partial (z+h(x))}{\partial x} \hat{i} + \frac{\partial (z+h(x))}{\partial z} \hat{k}$$

$$\nabla F = \frac{\partial h}{\partial x} \hat{i} + \hat{k} = \frac{\partial h}{\partial x} \hat{i} + \hat{k}$$

$$\mathbf{n} = \frac{\nabla F}{|\nabla F|} = \frac{\frac{dh}{dx} \hat{i} + \hat{k}}{\sqrt{1 + \left(\frac{\partial h}{\partial x}\right)^2}} \quad (8)$$

$$\mathbf{u} \cdot \mathbf{n} = u \frac{dh}{dx} + w = 0 \quad \text{on } z = -h(x) \quad (9.a)$$

$$\text{Or} \quad w = -u \frac{dh}{dx} \quad \text{on } z = -h(x) \quad (9.b)$$

Special case of horizontal bottom $\frac{\partial h}{\partial x} = 0$

$$\Rightarrow w = 0$$

For sloping bottom

$$\frac{w}{u} = -\frac{dh}{dx} \quad (10)$$

- Question?? Could we treat bottom as a streamline?

Yes since flow is everywhere tangential to it

- Bottom BC (Eq. 7) also applies to flows in 3D in which h is $h(x,y)$

➤ **Kinematic Free surface Boundary Condition (KFSBC)**

- **Free surface of wave is written as**

$$F(x, y, z, t) = z - \eta(x, y, t) = 0$$

Here $\eta(x, y, t)$ is displacement of free surface about horizontal plane $z=0$

$$\nabla F = \frac{\partial F}{\partial x} \hat{i} + \frac{\partial F}{\partial y} \hat{j} + \frac{\partial F}{\partial z} \hat{k} = \frac{-\partial \eta}{\partial x} \hat{i} - \frac{\partial \eta}{\partial y} \hat{j} + \hat{k}$$

$$|\nabla F| = \sqrt{1 + \left(\frac{\partial \eta}{\partial x}\right)^2 + \left(\frac{\partial \eta}{\partial y}\right)^2} \quad \text{and} \quad \mathbf{n} = \frac{\nabla F}{|\nabla F|}$$

$$\mathbf{U} = u\hat{i} + v\hat{j} + w\hat{k}$$

$$\mathbf{u} \cdot \mathbf{n} = \frac{-u \frac{\partial \eta}{\partial x} - v \frac{\partial \eta}{\partial y} + w}{\sqrt{1 + \left(\frac{\partial \eta}{\partial x}\right)^2 + \left(\frac{\partial \eta}{\partial y}\right)^2}} \quad (11.a)$$

Using eqn (6) $u \cdot n = \frac{-\frac{\partial F}{\partial t}}{|\nabla F|}$

$$\frac{-u \frac{\partial \eta}{\partial x} - v \frac{\partial \eta}{\partial y} + w}{\sqrt{1 + \left(\frac{\partial \eta}{\partial x}\right)^2 + \left(\frac{\partial \eta}{\partial y}\right)^2}} = \frac{\frac{\partial \eta}{\partial t}}{\sqrt{1 + \left(\frac{\partial \eta}{\partial x}\right)^2 + \left(\frac{\partial \eta}{\partial y}\right)^2}}$$

$$\Rightarrow w = \frac{\partial \eta}{\partial t} + u \frac{\partial \eta}{\partial x} + v \frac{\partial \eta}{\partial y} \quad (11.b)$$

$$z = \eta(x, y, t)$$

Dynamic Free surface Boundary Condition (DFSBC)

- **BC for fixed surfaces are easy to prescribe as they are applied on known surface.**

The displacement of upper boundary in free surface problem is not known a priori in water wave problem.

- **Fixed surface can support pressure variations across interface whereas free surface cannot.**

- Another BC is hence required on any free surface or interface to prescribe pressure distribution on the boundary. This is called Dynamic Boundary Condition.
- The DFSBC requires that the pressure on free surface be uniform along the wave form. Thus, unsteady Bernoulli's equation is applied on free surface $Z = \eta(x, t)$.

$$-\frac{\partial \phi}{\partial t} + \frac{1}{2}(u^2 + w^2) + \frac{p_n}{\rho} + gz = c(t) \quad (12)$$

p_n is constant and usually taken as gauge pressure $p_n = 0$

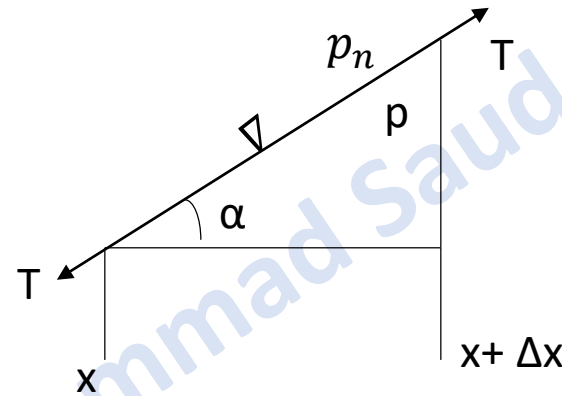
Class problem

If wave length are very short (order of several cm) that the surface is no longer “free”. Write DFSBC for this case where surface tension T is important.

Use $\sigma' = \frac{T}{l}$ (surface tension per unit length).

Solution

consider a surface for which curvature exists as shown below



Denoting p as the pressure under the free surface a free body analysis in vertical direction gives

$$\tau \left[-\sin \alpha \Big|_x + \sin \alpha \Big|_{x+\Delta x} \right] + (p - p_n) \Delta x + \text{terms of } \Delta x^2 = 0$$

$$\text{Here } \frac{\partial \eta}{\partial x} \approx \sin \alpha$$

Expanding using Taylor series and allow $\Delta x \longrightarrow 0$

$$P = p_n - \rho \frac{\partial^2 \eta}{\partial x^2}$$

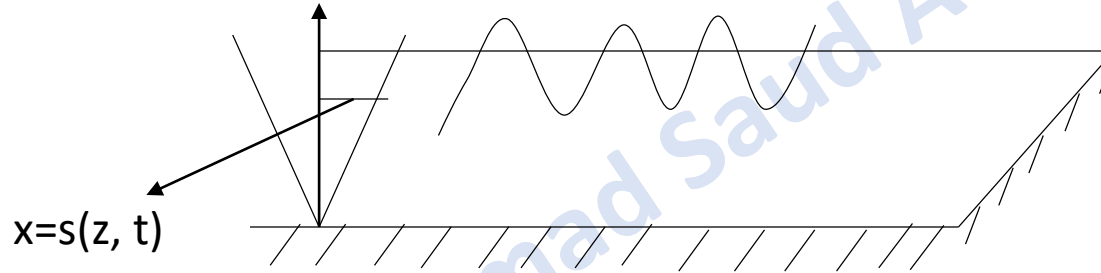
Hence DFSBC becomes

$$-\frac{\partial \phi}{\partial t} + \frac{1}{2} \left[\left(\frac{\partial \phi}{\partial x} \right)^2 + \left(\frac{\partial \phi}{\partial z} \right)^2 \right] + \frac{p_n}{\rho} - \frac{\rho}{\rho} \frac{\partial^2 \eta}{\partial x^2} + gz = c(t) \quad \text{at } z = \eta(x, t)$$

Lateral Boundary Conditions

- Until now we have discussed BC for bottom and upper surfaces.
- Conditions must also be specified on remaining lateral boundaries.
- If waves are propagating in x directions $\Rightarrow$ no flow in y directions is lateral BC.
- In x direction if motion occurs due to a paddle or wave maker then usual kinematic BC applied.

Consider a vertical paddle. If displacement of paddle is described as $x=s(z, t)$ find KBC.



➤ For waves that are periodic in space and time the BC is expressed as

$$\phi(x,t) = \phi(x+L,t) \quad (13.a)$$

$$\phi(x,t) = \phi(x,t+T) \quad (13.b)$$

Velocity potential derivation assumptions

➤ **The assumptions in deriving the expression for the velocity potential due to propagating ocean waves are;**

- **Flow is said to be irrotational**
- **Fluid is ideal**
- **Surface tension is neglected**
- **Pressure at the free surface is uniform and constant**
- **The seabed is rigid, horizontal and impermeable**

Velocity potential derivation assumptions

- Wave height is small compared to its length
- Potential flow theory is applicable
- A velocity potential ϕ exists and the velocity components u and w in the x and z directions can be obtained as $\frac{\partial \phi}{\partial x}$ and $\frac{\partial \phi}{\partial z}$.

DERIVATION FOR VELOCITY POTENTIAL

The governing equation is the Laplace Equation given by

$$\nabla^2 \phi = 0 \quad (2.1)$$

The continuity equation and Bernoulli's equations (2.2) and (2.3) are used in the solution procedure

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (2.2)$$

$$-\frac{\partial \phi}{\partial t} + \frac{1}{2}(u^2 + v^2 + w^2) + \frac{p}{\rho} + gz = 0 \quad (2.3)$$

➤ **BOUNDARY CONDITIONS:**

- The equation (2.1) is to be satisfied in the region $-d \leq z \leq \eta$, $-\infty \leq x \leq \infty$ where η is the water surface elevation measured from the Still Water Level (SWL).
- The kinematic bottom boundary condition meaning, that the vertical velocity component at the sea bottom is zero. Since 'z' is negative in downward from SWL.
- The pressure at the free surface is zero or at $z = \eta$

Linearizing the Bernoulli's equation results in

$$-\frac{\partial \phi}{\partial t} + \frac{p}{\rho} + gz = 0 \quad (2.4)$$

- When $z = \eta$ and taking $p = 0$ using equation (2.4) we get

$$\eta = \frac{1}{g} \left[\frac{\partial \phi}{\partial t} \right]_{z=\eta}$$

- This is the dynamic free surface boundary condition. Since we assume that amplitude of the waves are small, the above equation can be written as

$$\eta = \frac{1}{g} \left[\frac{\partial \phi}{\partial t} \right]_{z=0} \quad (2.5)$$

This is applicable only when η is small and valid for $\frac{H}{d}$ and $\frac{H}{L} < 1$. The definition sketch is given in Fig.2.2. With the above boundary conditions, the solution to eqn. (2.1) is solved

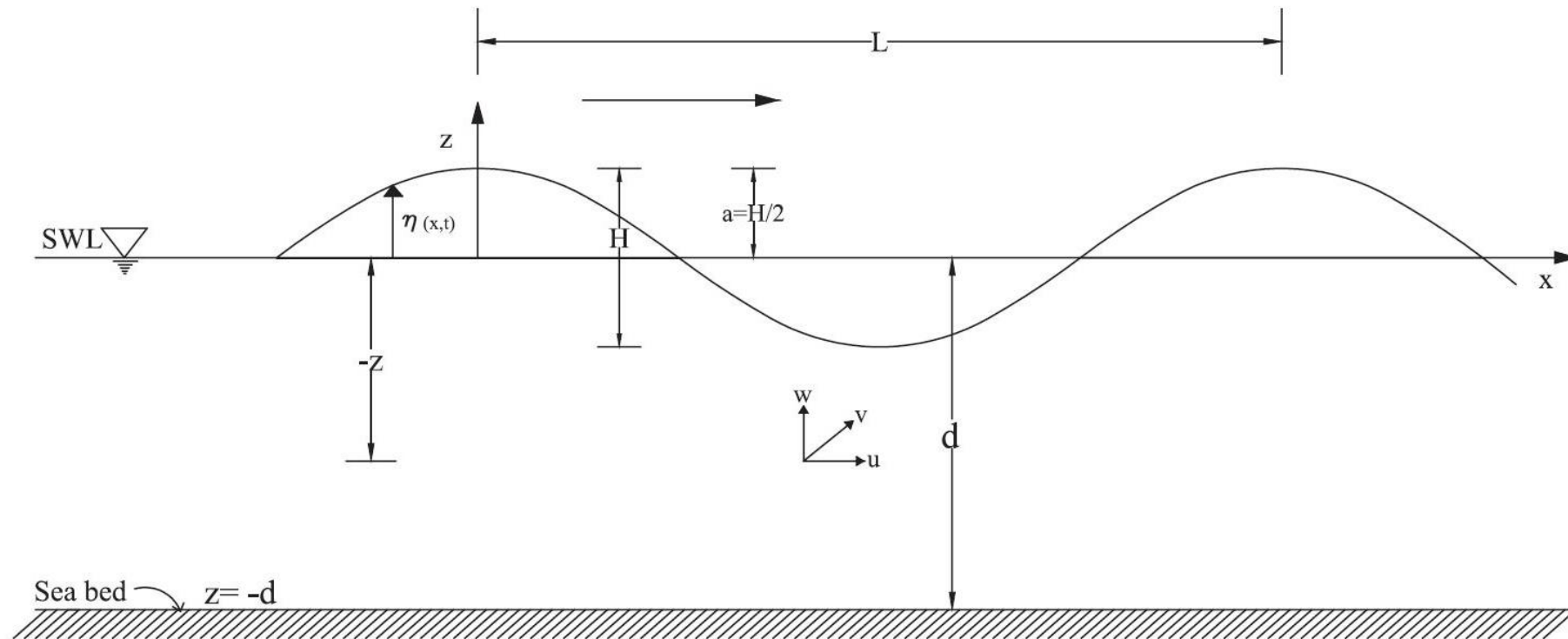


Fig. 2.2 Definition sketch for wave motion

SOLUTION TO THE LAPLACE EQUATION:

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad (2.6)$$

- Method of separable is used to obtain the solution to Eqn. (2.6)
- Let us assume

$$\phi(x, z, t) = \bar{X}(x)\bar{Z}(z)\bar{T}(t) \quad (2.7)$$

- Substituting Eq. (2.7) in Eq. (2.6) we get

$$\overline{X''} \overline{Z} \overline{T} + \overline{X} \overline{Z''} \overline{T} = 0$$

- Where each prime denotes differentiation once with respect to the particular independent variable.
- Dividing both side of the above $\overline{X} \overline{Z} \overline{T}$ gives

$$\frac{\overline{X''}}{\overline{X}} = \frac{-\overline{Z''}}{\overline{Z}}$$

- Let this be constant = $-k^2$; then

$$\overline{X}'' + k^2 \overline{X} = 0 \quad (2.8)$$

$$\overline{Z}'' - k^2 \overline{Z} = 0 \quad (2.9)$$

$$\overline{X} = A \cos kx + B \sin kx$$

$$\overline{Z} = C e^{kz} + D e^{-kz}$$

Hence $\phi(x, z, t) = (A \cos kx + B \sin kx) (C e^{kz} + D e^{-kz}) \overline{T}(t)$

- The solutions to ϕ are simple harmonic in time requiring $\bar{T}(t)$ be replaced as $\cos(\sigma t)$ or $\sin(\sigma t)$, thus leading to four forms of solutions to ϕ , such that
- $\phi_1 = A_1 (C e^{kz} + D e^{-kz}) \cos(kx) \cdot \cos(\sigma t)$
- $\phi_2 = A_2 (C e^{kz} + D e^{-kz}) \sin(kx) \cdot \sin(\sigma t)$
- $\phi_3 = A_3 (C e^{kz} + D e^{-kz}) \sin(kx) \cdot \cos(\sigma t)$
- $\phi_4 = A_4 (C e^{kz} + D e^{-kz}) \cos(kx) \cdot \sin(\sigma t)$

DETERMINATION OF THE CONSTANTS:

The constants are determined by using the dynamic free surface boundary condition and the kinematic bottom boundary condition.

Considering ϕ_2

$$\phi_2 = A_2 (C e^{kz} + D e^{-kz}) \sin(kx) \cdot \sin(\sigma t) \quad (2.10)$$

- Applying the kinematic bottom boundary condition

$$\text{i.e., } \frac{\partial \phi}{\partial z} = 0 \text{ at } z = -d$$

$$\left. \frac{\partial \phi_2}{\partial z} \right|_{z=-d} = A_2 (Cke^{-kd} - Dke^{kd}) \sin(kx) \cdot \sin(\sigma t) = 0$$

$$A_2 \neq 0; \sin(kx) \cdot \sin(\sigma t) \neq 0 \text{ [since velocity potential exists]}$$

$$\therefore C = De^{2kd}$$

Substituting for C in (Eq. 2.10) and simplifying,

$$\phi_2 = 2A_2 D e^{kd} \left(\frac{e^{k(d+z)} + e^{-k(d+z)}}{2} \right) \sin(kx) \cdot \sin(\sigma t)$$

$$\phi_2 = 2A_2 D e^{kd} \cosh k(d+z) \sin(kx) \cdot \sin(\sigma t) \quad (2.11)$$

- $\frac{\partial \phi_2}{\partial t} \Big|_{z=0} = 2A_2 D \sigma e^{kd} \cdot \cosh kd \cdot \sin(kx) \cdot \cos(\sigma t)$

On assuming

- $\eta = a \sin(kx) \cdot \cos(\sigma t)$ where a = wave amplitude = $\frac{H}{2}$ and applying the free

surface boundary condition $\left\{ \eta = \frac{1}{g} \frac{\partial \phi_2}{\partial t} \Big|_{z=0} \right\}$ we get

- $a \sin(kx) \cdot \cos(\sigma t) = \frac{2A_2 D \sigma e^{kd}}{g} \cdot \cosh kd \cdot \sin(kx) \cdot \cos(\sigma t)$

$$2A_2 D e^{kd} = \frac{ag}{\sigma} \frac{1}{\cosh kd}.$$

Substituting in eq. (2.11), we get

$$\phi_2 = \frac{ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} \sin(kx) \cdot \sin(\sigma t) \quad (2.12)$$

Let us consider ϕ_3

$$\phi_3 = A_3 (C e^{kz} + D e^{-kz}) \sin(kx) \cdot \cos(\sigma t) \quad (2.13)$$

- Applying the kinematic bottom boundary condition

$$\frac{\partial \phi_3}{\partial z} \Big|_{z=-d} = A_3 (C k e^{-kd} - D k e^{kd}) \sin(kx) \cdot \cos(\sigma t) = 0$$

- $A_3 \neq 0$; $\sin(kx) \cdot \cos(\sigma t) \neq 0$

$$C = D e^{2kd}$$

- Substituting for C in eq. (2.13)

$$\phi_3 = 2A_3 D e^{kd} \cosh k(d+z) \sin(kx) \cdot \cos(\sigma t) \quad (2.14)$$

And

- $\left. \frac{\partial \phi_3}{\partial t} \right|_{z=0} = -2A_3 D \sigma e^{kd} \cdot \cosh kd \cdot \sin(kx) \cdot \sin(\sigma t)$
- On assuming $\eta = a \sin(kx) \cdot \sin(\sigma t)$ and applying the free surface boundary condition we get

$$2A_3 D e^{kd} = \frac{-ag}{\sigma} \frac{1}{\cosh kd}.$$

Substituting in eq. (2.14), we get

$$\phi_3 = \frac{-ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} \sin(kx) \cdot \cos(\sigma t) \quad (2.15)$$

Let us consider ϕ_4

$$\phi_4 = A_4 (C e^{kz} + D e^{-kz}) \cos(kx) \cdot \sin(\sigma t) \quad (2.16)$$

Applying the kinematic bottom boundary condition

$$\left. \frac{\partial \phi_4}{\partial z} \right|_{z=-d} = A_4 (C k e^{-kd} - D k e^{kd}) \cos(kx) \cdot \sin(\sigma t) \longrightarrow C = D e^{2kd}$$

Substituting for C in eq. (2.16)

- $\phi_4 = 2A_4 D e^{kd} \cosh k(d+z) \cos(kx) \cdot \sin(\sigma t)$ (2.17)

And

- $\left. \frac{\partial \phi_4}{\partial t} \right|_{z=0} = 2A_4 D e^{kd} \cosh k(d+z) \cos(kx) \cdot \sin(\sigma t)$

Assuming $\eta = a \cos(kx) \cdot \cos(\sigma t)$ and applying eq (2.5)

We get $2A_4 D e^{kd} = \frac{ag}{\sigma} \frac{1}{\cosh kd}$.

- Substituting in eq. (2.17), we get

$$\phi_4 = \frac{ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} \cos(kx) \cdot \sin(\sigma t) \quad (2.18)$$

- Let us consider ϕ_1

$$\phi_1 = A_1 (C e^{kz} + D e^{-kz}) \cos(kx) \cdot \cos(\sigma t) \quad (2.19)$$

- Applying the kinematic bottom boundary condition

$$\left. \frac{\partial \phi_1}{\partial z} \right|_{z=-d} = 0 \quad C = D e^{2kd}$$

- Substituting for C in eq. (2.19)
- $\phi_1 = 2A_1 D e^{kd} \cosh k(d+z) \cos(kx) \cdot \cos(\sigma t)$ (2.20)
- assuming $\eta = a \cos(kx) \cdot \sin(\sigma t)$ and applying equation (2.5)

We get

- $2A_1 D e^{kd} = \frac{-ag}{\sigma} \frac{1}{\cosh kd}$
- Substituting in eq. (2.20), we get
- $\phi_1 = \frac{-ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} \cos(kx) \cdot \cos(\sigma t)$ (2.21)

- If $\phi^+ = \phi_2 - \phi_1$

$$= \frac{ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} [\cos(kx) \cdot \cos(\sigma t) + \sin(kx) \cdot \sin(\sigma t)]$$

$$\phi = \frac{ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} \cdot \cos(kx - \sigma t). \quad (2.22)$$

- This is the expression for the velocity potential for a propagating wave in a constant water depth

Since $\eta = \frac{1}{g} \frac{\partial \phi}{\partial t} \Big|_{z=0}$

$$\eta = \frac{1}{g} \frac{ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} \cdot \sigma \sin(kx - \sigma t).$$

Hence $\eta = a \sin(kx - \sigma t).$ (2.23)

- ‘ η ’ is periodic in x and t . If we locate a point and traverse along the wave, such that, at all-time ‘ t ’ our position relative to the wave form remains fixed then the phase difference is zero or $kx - \sigma t = \text{constant}$

- And the speed with which we must move to accomplish this is given by $kx = \sigma t + \text{constant}$

$$k \frac{dx}{dt} = \sigma$$

$$\frac{dx}{dt} = \frac{\sigma}{k} = \frac{2\pi}{T} \frac{L}{2\pi} = \frac{L}{T} = C$$

- $C = \frac{L}{T} = \text{CELERITY or Speed of the wave} \quad (2.24)$

Wave moving in negative 'x' direction

- If $\phi^- = \phi_2 + \phi_1$

$$= \frac{-ag}{\sigma} \frac{\cosh k(kx + \sigma t)}{\cosh kd} \cdot \cos(kx + \sigma t).$$

- Since $\eta = \frac{1}{g} \frac{\partial \phi}{\partial t} \Big|_{z=0}$

$$\eta = \frac{1}{g} \frac{-ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} (-\sigma \sin(kx + \sigma t))$$

$$\eta = a \sin(kx + \sigma t)$$

To obtain the celerity of the wave we have

$$kx + \sigma t = \text{constant}$$

$$\frac{dx}{dt} = \frac{-\sigma}{k} = \frac{-2\pi}{T} \frac{L}{2\pi} = \frac{-L}{T} = -C \quad (2.25)$$

DISPERSION RELATIONSHIP

- The relationship between wavelength, period and water depth is obtained as given below. The main assumption while establishing the relationship is that, since, we are dealing with small amplitude waves, meaning that the slope of the wave profile are small so that $\frac{d\eta}{dt}$ can be approximately said as equal to the vertical component velocity, w . This is

- $$w = \frac{d\eta}{dt} = \frac{\partial \eta}{\partial t} + \frac{\partial \eta}{\partial x} \cdot \frac{\partial x}{\partial t}$$

- Wave slope being small by setting, $\frac{\partial \eta}{\partial x} = 0$

$$w = \frac{\partial \eta}{\partial t} \text{ but } w = \frac{-\partial \phi}{\partial z}$$

$$\text{Hence } \frac{\partial \eta}{\partial t} = \frac{-\partial \phi}{\partial z} \quad (2.26)$$

$$\text{Differentiating the expression of } \eta \text{ we get } \frac{\partial \eta}{\partial t} = \frac{1}{g} \frac{\partial^2 \phi}{\partial t^2} \Big|_{z=0}$$

$$\text{Hence } \frac{\partial \eta}{\partial t} = \frac{-A\sigma^2}{g} \cosh kd \cdot \cos(kx - \sigma t). \quad (2.27)$$

- Where $A = \frac{H}{2} \frac{g}{\sigma} \frac{1}{\cosh kd}$

$$w = \frac{-\partial \phi}{\partial z} = -Ak. \sinh kd. \cos(kx - \sigma t). \quad (2.28)$$

- Using the relation of Eq. (2.26), equating Eq. (2.27) to Eq (2.28), we get

$$\frac{A\sigma^2}{g} \cosh kd. \cos(kx - \sigma t) = Ak. \sinh kd. \cos(kx - \sigma t).$$

- $\frac{\sigma^2}{g} = \frac{k \sinh kd}{\cosh kd}$

- $\sigma^2 = gk \cdot \tanh kd$ (2.29)

- σ : Wave angular frequency = $\frac{2\pi}{T}$ and k : wave number = $\frac{2\pi}{L}$

- The above equation can be written as

- $\left(\frac{2\pi}{T}\right)^2 = g \left(\frac{2\pi}{L}\right) \tanh kd$

$$\left(\frac{L}{T}\right)^2 = \frac{gL}{2\pi} \tanh kd$$

$$(C)^2 = \frac{g}{k} \tanh kd \quad (2.30)$$

The speed at which a wave moves in its direction of propagation as a function of water depth is given by Eq.(2.30)

Since

$C = \frac{L}{T}$ from the above equation we get

- $C = \sqrt{\frac{gL}{2\pi} \tanh kd}$ (2.31)

Or $L = \frac{gT^2}{2\pi} \tanh kd$ (2.32)

- Since the unknown 'L' occurs on both sides (Implicit Eq.) of Eq. (2.32), it has to be solved by trial and error.

CELERITY IN DIFFERENT WATER DEPTH CONDITIONS:

Classification	d/L	$2\pi d/L$	$\tanh(2\pi d/L)$
Deep Waters	$>1/2$	$>\pi$	~ 1
Intermediate waters	$1/20$ to $1/2$	$\pi/10$ to π	$\tanh(2\pi d/L)$
Shallow waters	$\leq 1/20$	0 to $\pi/10$	$\sim 2\pi d/L$

Classification of ocean waves according to water depth

Deep water conditions:

- In case of deep waters Eq (2.30) becomes

$$c_0 = \sqrt{\frac{gL_0}{2\pi}} \quad \text{since, } \tanh kd = 1$$

and Eq (2.31) becomes

$$c_0 = \frac{gT}{2\pi} \quad (2.33)$$

Or
$$L_0 = \frac{gT^2}{2\pi} \quad (2.34)$$

That is when $d/L \geq 1/2$, $\tanh(kd)$ approaches unity and the wave characteristics are independent of the water depth, d , while wave period remaining constant. Hence,

$$L_0 = \frac{gT^2}{2\pi} = 5.12 T^2 \quad [\text{FPS}]$$

$$L_0 = \frac{gT^2}{2\pi} = 1.56 T^2 \quad [\text{MKS}]$$

- If Eq (2.33) or (2.34) are used to compute wave celerity for shallow water conditions ($d/L < 1/20$) and error of about 20% to 50% results.

Shallow water conditions:

- When $kd = \frac{\pi}{10}$, $\frac{d}{L} \leq \frac{1}{20}$

$$C^2 = \frac{gL}{2\pi} \cdot \tanh(kd)$$

- Hence $\tanh(kd) \approx kd = 2\pi d/L$ $C^2 = \frac{gL}{2\pi} \cdot \frac{2\pi d}{L}$ $C^2 = gd$

$$C = \sqrt{gd} \quad (2.35)$$

This relation shows that when a wave travels in shallow waters wave celerity depends only on the water depth.

Relationship between d/L and d/L_0

- It can be shown by dividing the Eq.(2.31) by Eq.(2.33) and dividing Eq.

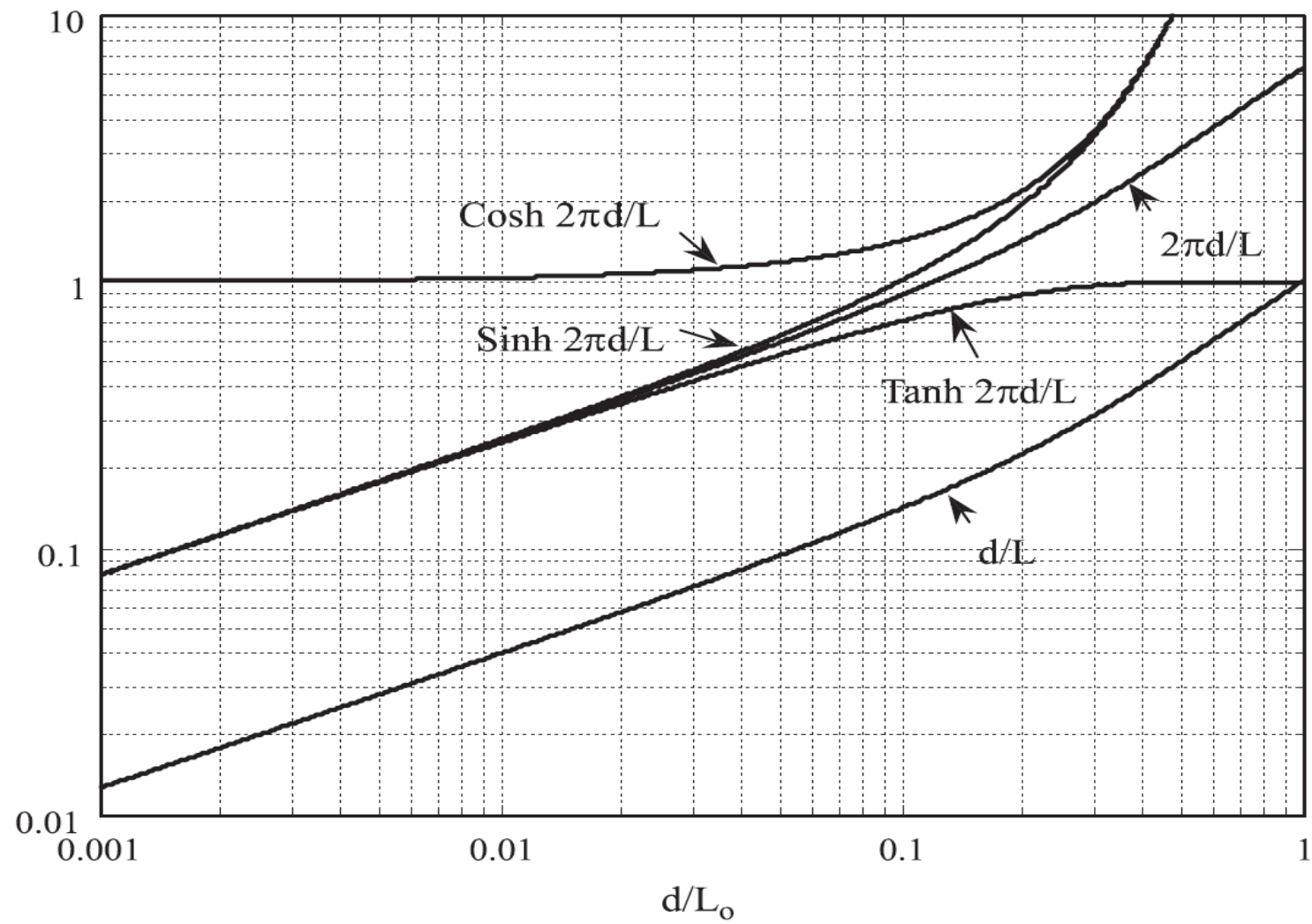
(2.32) by Eq.(2.34) that $C/C_0 = L/L_0 = \tanh kd$

Multiplying both sides by d/L , then

$$d/L_0 = d/L \tanh kd \quad (2.36)$$

The relation between d/L and d/L_0 is given in the wave Tables

Relationship between d/L_0 and functions of d/L



LOCAL FLUID PARTICLE VELOCITIES AND ACCELERATION UNDER PROGRESSIVE WAVES

- In the evaluation of wave forces on offshore structures it is desirable to know the fluid particle kinematics that is velocity and acceleration.

We know

$$\phi = \frac{ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} \cos(kx - \sigma t) \quad (2.37)$$

- The horizontal water particle velocity or orbital velocity u is given by

$$u = -\frac{\partial \phi}{\partial x} = \frac{ag}{\sigma} k \frac{\cosh k(d+z)}{\cosh kd} \sin(kx - \sigma t) \quad (2.38)$$

- The horizontal water particle velocity or orbital velocity, u is given by

$$= \frac{ag}{\sigma} k \frac{\cosh k(d+z)}{\sinh kd} \cdot \tanh kd \cdot \sin(kx - \sigma t)$$

- Substituting the relationship $C^2 = \frac{g}{k} \tanh kd$ (2.30) and $a = \frac{H}{2}$

In the above expression we get

$$U = \frac{H}{2\sigma} C^2 k^2 \frac{\cosh k(d+z)}{\sinh kd} \sin(kx - \sigma t)$$

$$= \frac{H}{2} \left(\frac{L}{T} \right)^2 \frac{(2\pi/L)^2}{(2\pi/T)} \frac{\cosh k(d+z)}{\sinh kd} \sin(kx - \sigma t)$$

Simplifying we get,

$$U = \frac{\pi H}{T} \frac{\cosh k(d+z)}{\sinh kd} \sin(kx - \sigma t) \quad (2.39)$$

The vertical fluid particle velocity, w is given by

$$w = -\frac{\partial \phi}{\partial z} = \frac{-ag}{\sigma} k \frac{\sinh k(d+z)}{\cosh kd} \cos(kx - \sigma t) \quad (2.40)$$

$$= \frac{-ag}{\sigma} k \frac{\sinh k(d+z)}{\sinh kd} \tanh kd \cdot \cos(kx - \sigma t)$$

Using eq (2.30) we get

$$w = \frac{-\pi H}{T} \frac{\sinh k(d+z)}{\sinh kd} \cos(kx - \sigma t)$$

$$w = \frac{-\pi H}{T} \frac{\sinh k(d+z)}{\sinh kd} \cos(kx - \sigma t)$$

(2.41)

The equations express the velocity components with the wave at any depth z . At a given z , all the velocities are seen to be harmonic in x and t as shown in 2.4.a

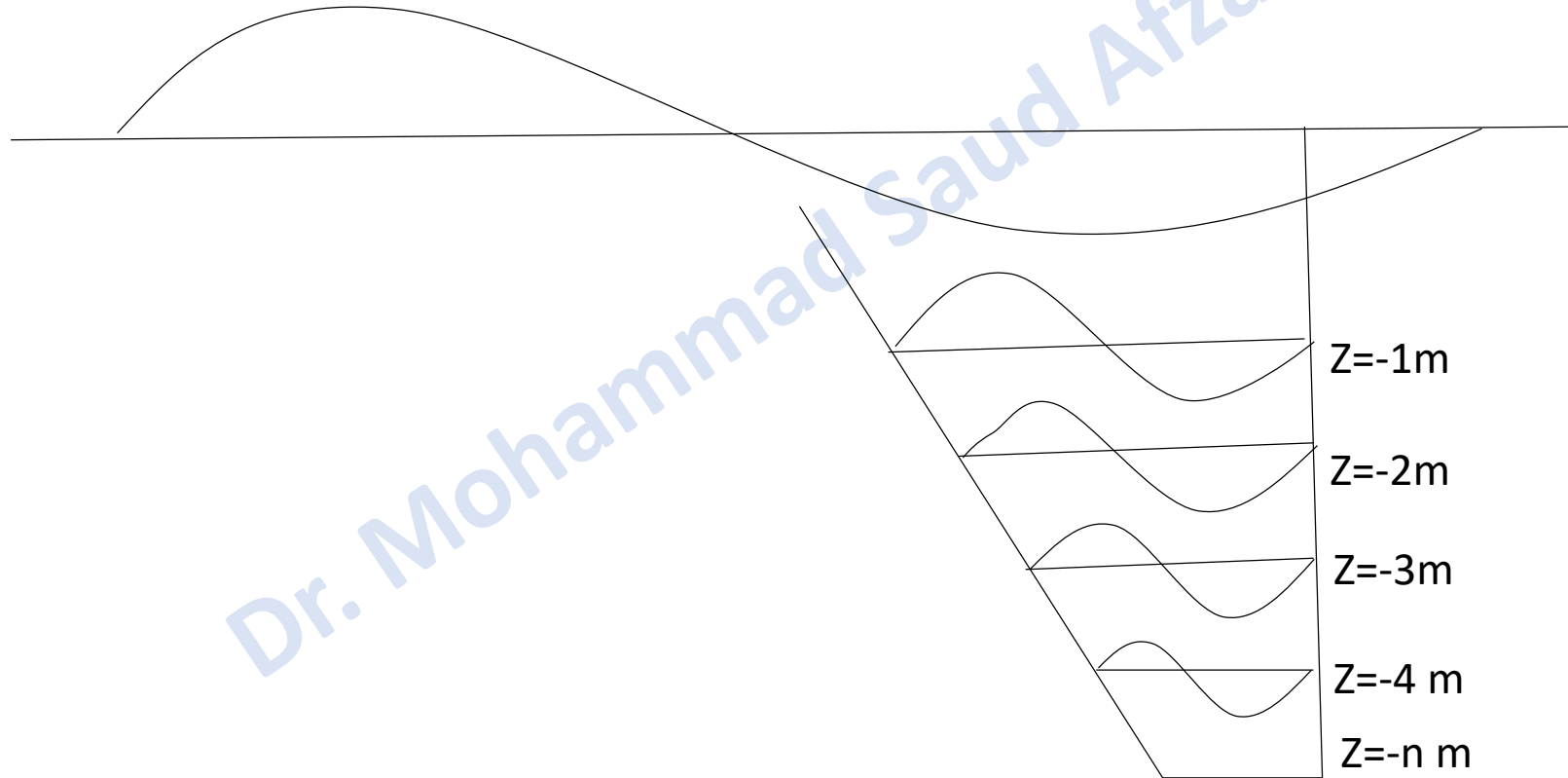


Fig.2.4a shows the horizontal velocity components are the hyperbolic functions of depth

At a given phase angle $\theta, (\theta=kx-\sigma t)$ the hyperbolic function of z (cosh and sinh) cause an exponential decay of u and w with the distance down from free surface. This is indicated schematically in fig 2.4.b for the phase angles at which the components are largest. The variation of u and w with respect to phase are shown in fig 2.5.

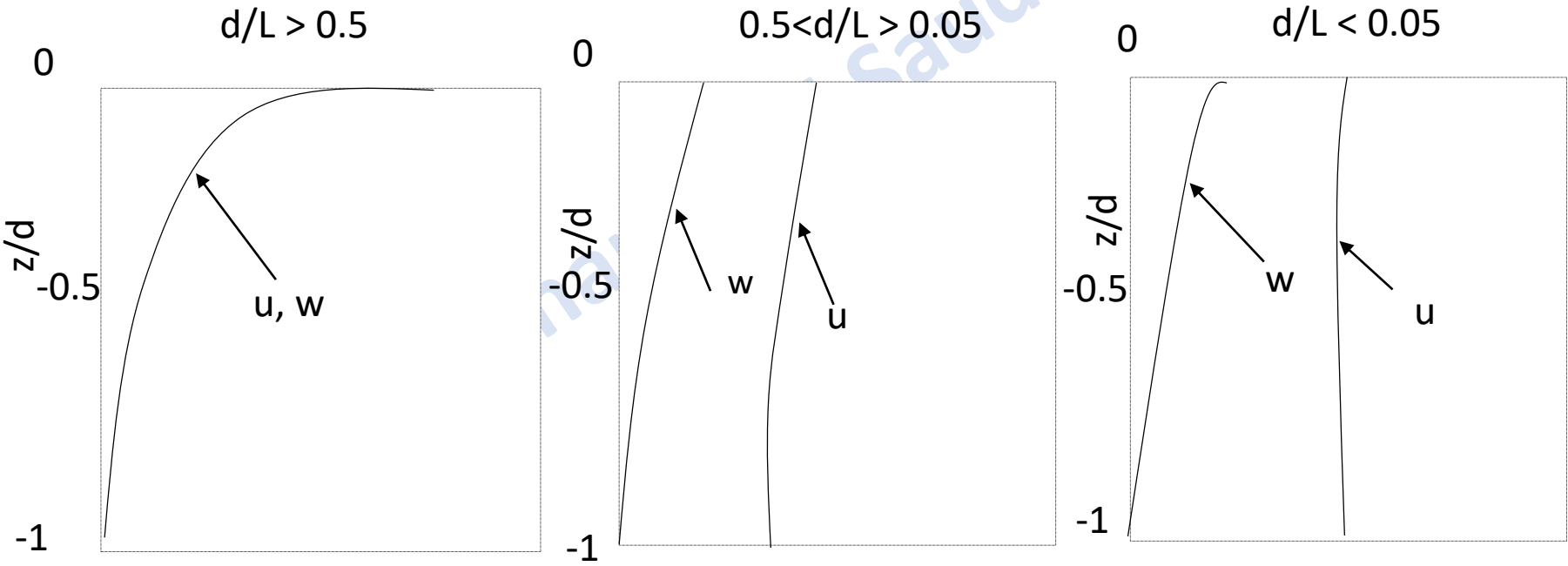


Fig 2.4.b Variation of maximum u and w

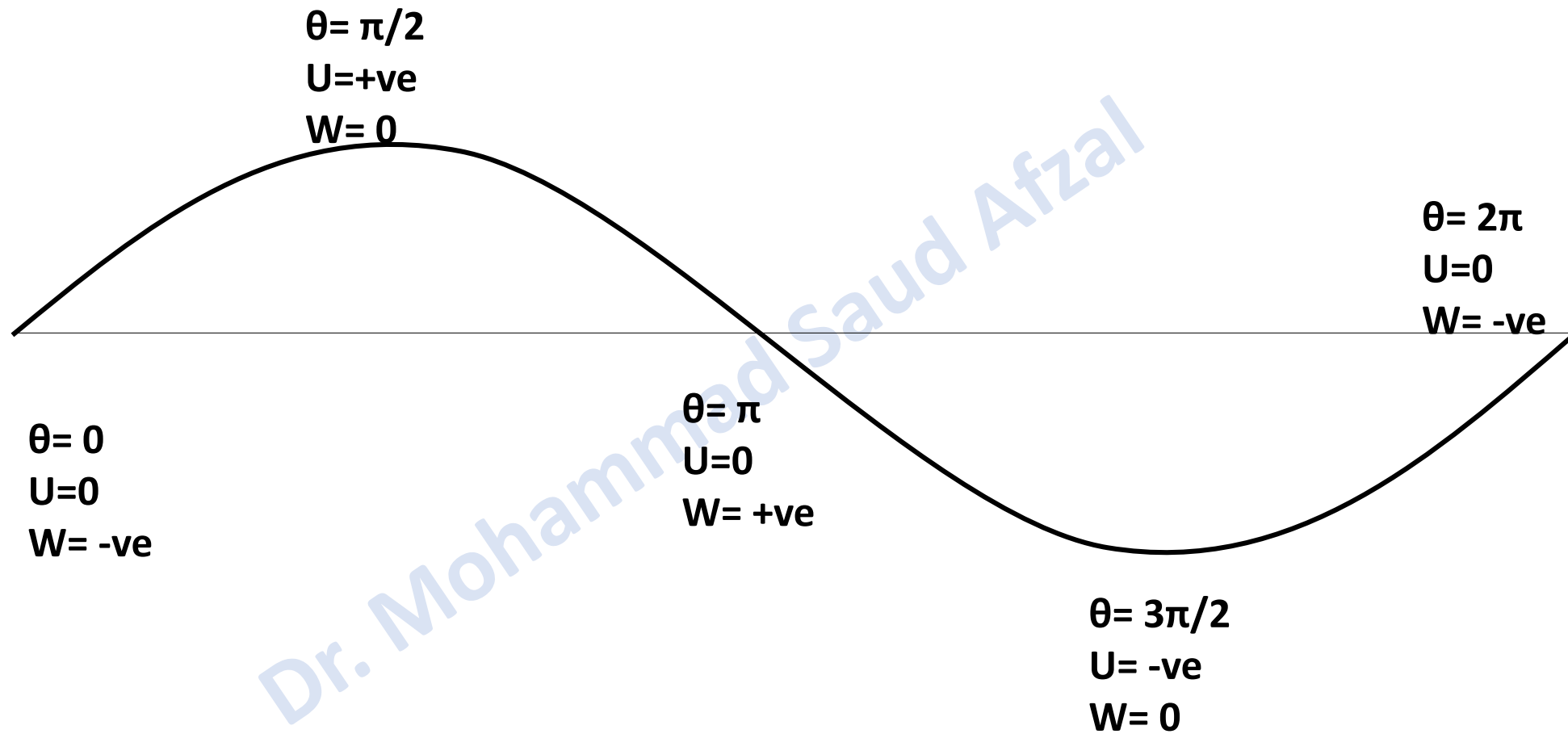


Fig 2.5 Variation of u and w with phase

- The local acceleration in x and z direction are given by

$$a_x = \frac{\partial u}{\partial t} = \frac{-2\pi^2 H \cosh k(d+z)}{T^2 \sinh kd} \cos(kx - \sigma t) \quad (2.42)$$

$$a_w = \frac{\partial w}{\partial t} = \frac{-2\pi^2 H \sinh k(d+z)}{T^2 \sinh kd} \sin(kx - \sigma t) \quad (2.43)$$

- This expression for the fluid particle kinematics reported above is for the water surface elevation, η being a sinus curve. The fluid particle kinematics for η being a cosine curve are given below.

The ϕ can be derived as

$$\phi = \frac{ag}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} \cdot \sin(kx - \sigma t). \quad (2.44)$$

In which case

$$\eta = a \cos(kx - \sigma t) \quad (2.45)$$

$$u = \frac{agk}{\sigma} \frac{\cosh k(d+z)}{\cosh kd} \cos(kx - \sigma t) \quad (2.46)$$

$$a_x = agk \frac{\cosh k(d+z)}{\cosh kd} \sin(kx - \sigma t) \quad (2.47)$$

$$w = \frac{ag}{\sigma} k \frac{\sinh k(d+z)}{\cosh kd} \sin(kx - \sigma t) \quad (2.48)$$

$$a_w = -agk \frac{\sinh k(d+z)}{\cosh kd} \cos(kx - \sigma t) \quad (2.49)$$

WATER PARTICLE DISPLACEMENT UNDER PROGRESSIVE WAVE

- The expression for individual horizontal and vertical water particle displacements is Obtained as follows.

$$\delta_x = \int u dt = \frac{H}{2} \frac{\cosh k(d+z)}{\sinh kd} \cos(kx - \sigma t) \quad (2.50)$$

$$\delta_z = \int w dt = \frac{H}{2} \frac{\sinh k(d+z)}{\sinh kd} \sin(kx - \sigma t) \quad (2.51)$$

Let $\delta_x = D \cos(kx - \sigma t)$ where $D = \frac{H}{2} \frac{\cosh k(d+z)}{\sinh kd}$

$$\delta_z = B \sin(kx - \sigma t) \text{ where } B = \frac{H}{2} \frac{\sinh k(d+z)}{\sinh kd}$$

$$\cos^2(kx - \sigma t) = \left(\frac{\delta_x}{D}\right)^2 : \sin^2(kx - \sigma t) = \left(\frac{\delta_z}{B}\right)^2$$

Since, $[\cos^2(kx - \sigma t) + \sin^2(kx - \sigma t) = 1]$, we have

$$\boxed{\left(\frac{\delta_x}{D}\right)^2 + \left(\frac{\delta_z}{B}\right)^2 = 1} \quad (2.52)$$

This equation of an ellipse showing that the water particles moves in an elliptical orbit.

Where, D= Semi major axis (horizontal measure of particle displacement)

B= Semi minor axis (vertical measure of particle displacement)

➤ **Shallow water Condition:**

For $\frac{d}{L} < \frac{1}{20}$ we have used $\cosh k(d+z)$ and $\sinh k(d+z)$

$$\sinh k(d+z) \longrightarrow k(d+z)$$

$$\sinh kd \longrightarrow kd$$

$$\text{Hence, } D = \frac{H}{2} \cdot \frac{1}{kd}$$

$$B = \frac{H}{2} \frac{k(d+z)}{kd} = \frac{H}{2} \frac{(d+z)}{d}$$

- Hence, the water particles move in elliptical orbits (paths) in shallow and intermediate waters with the equation of the form

$$\boxed{\left(\frac{\delta_x}{\frac{H}{2kd}}\right)^2 + \left(\frac{\delta_z}{\frac{H(d+z)}{2d}}\right)^2 = 1} \longrightarrow (2.53)$$

➤ Deep water condition:

For the case $\frac{d}{L} > \frac{1}{2}$

$$D = \frac{H \cosh k(d+z)}{2 \sinh kd} = \frac{H}{2} \left(\frac{e^{k(d+z)} + e^{-k(d+z)}}{e^{kd} - e^{-kd}} \right)$$

As 'd' (depth of water of d/L) is very large $e^{-k(d+z)}$ and e^{-kd} will be very small compared to $e^{k(d+z)}$

Hence
$$D = \frac{H}{2} \left(\frac{e^{k(d+z)}}{e^{kd}} \right) = \frac{H}{2} e^{kz}$$

Similarly $B = \frac{H}{2} e^{kz}$

- Thus, the particles move in circular orbits in deep waters (since $D=B$) with the equation of the form

$$\boxed{\left(\frac{\delta_x}{\frac{H}{2} e^{kz}} \right)^2 + \left(\frac{\delta_z}{\frac{H}{2} e^{kz}} \right)^2 = 1} \longrightarrow (2.54)$$

This shows that for deep water conditions, the water particles are circular.

- The amplitude of the water particle displacement decreases exponentially along with the depth. The water particle displacements becomes small relative to the wave height at a depth equal to one half the wave length below the SWL. The variation of the water particle displacements under different depth conditions is illustrated in Fig 2.6

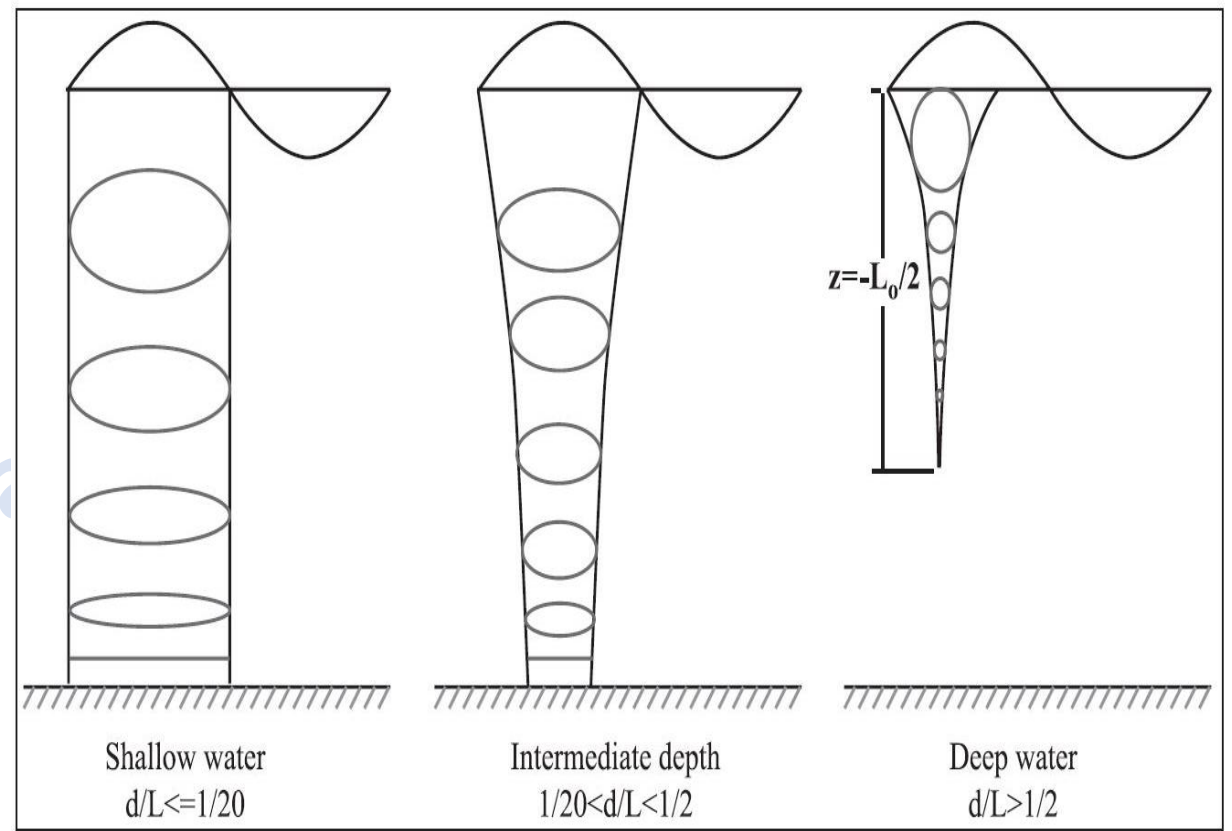


Fig. 2.6 Schematic representation of fluid particle trajectories

Solution to the Dispersion equation

- An approximate solution for wave number k in the dispersion relationship given by eq. (2.29)
- For a given σ and d proposed by Hunt (1979) can be solved directly for kd .

$$(kd)^2 = y^2 + \frac{y}{1 + \sum_{n=1}^6 d_n y^n} \quad (2.55)$$

Where $y = \frac{\sigma^2 d}{g} = k_0 d$ and

$$\begin{aligned} d_1 &= 0.666666666 & d_2 &= 0.355555555 & d_3 &= 0.160846508 \\ d_4 &= 0.0632098765 & d_5 &= 0.0217540484 & d_6 &= 0.0065407983 \end{aligned}$$

The celerity can be obtained as

$$\frac{c^2}{gd} = \left[y + (1 + 0.6522y + 0.4622y^2 + 0.0864y^4 + 0.0675y^5)^{-1} \right]^{-1} \quad (2.56)$$

Which is accurate to 0.1% for $0 < y < \infty$

PRESSURE DISTRIBUTION UNDER PROGRESSIVE WAVES:

- The linearized Bernoulli's equation is given by

$$\frac{-\partial\phi}{\partial t} + \frac{p}{\rho} + gz = 0$$

- Multiplying through out by ρ the total pressure is given as,

$$p = \rho \frac{\partial\phi}{\partial t} + (-\gamma z) \quad (\text{Dynamic} + \text{Static})$$

- Substituting for ϕ from eq. (2.22) we get

$$p = \frac{\gamma H}{2} \frac{\cosh k(d+z)}{\cosh kd} \sin(kx - \sigma t) - \gamma z$$

$$\eta = \frac{H}{2} \sin(kx - \sigma t) \text{ and let } \frac{\cosh k(d+z)}{\cosh kd} = K_p$$

Where K_p is the pressure response factor then,

$$P = \gamma \eta K_p - \gamma z$$

Or
$$\frac{P}{\gamma} = (\eta K_p - z) \quad (2.57)$$

It is to be mentioned that p was set to zero to define the free surface boundary condition in the Bernoulli equation. However ϕ was determined by setting $p=0$ at $z=0$ instead of $z=\eta$ [Refer Eq 2.5]. Hence eq 2.57 is valid only for negative z .

Applying Eq (2.57)

Pressure at $z=0$, $\frac{P}{\gamma} = \eta$

Pressure at $z=-d$, $\frac{P}{\gamma} = \frac{\eta}{\cosh kd} + d$ (2.58)

This is $[\frac{\eta}{\cosh kd} + d] < d + \eta$

Since $\cosh kd$ is always greater than 1

Under the trough at sea bed

Conditions are $z = -d$, $\eta = -\eta$

Substituting $K_p = \frac{\cosh(d-d)}{\cosh kd} = \frac{1}{\cosh kd}$

And $\eta = -\eta$ Hence from eq (2.57)

$$\frac{P}{\gamma} = \frac{-\eta}{\cosh kd} + d \quad (2.59)$$

$$\frac{P}{\gamma} = \left[d - \frac{\eta}{\cosh kd} \right] > (d - \eta)$$

It is often needed to determine the surface wave height based on subsurface measurement of pressure. For this purpose Eq. (2.57) is represented as

$$\eta = \frac{N(p + \rho g z)}{\rho g K}, \text{ where 'K' is pressure response factor at the seabed given by } \frac{1}{\cosh kd}.$$

N is the correction factor depending on the period, depth, wave amplitude etc.

$N > 1$ for long period waves

$N < 1$ for short period waves

$N = 1$ for linear waves

The pressure distribution under a progressive wave is given in Fig 2.7

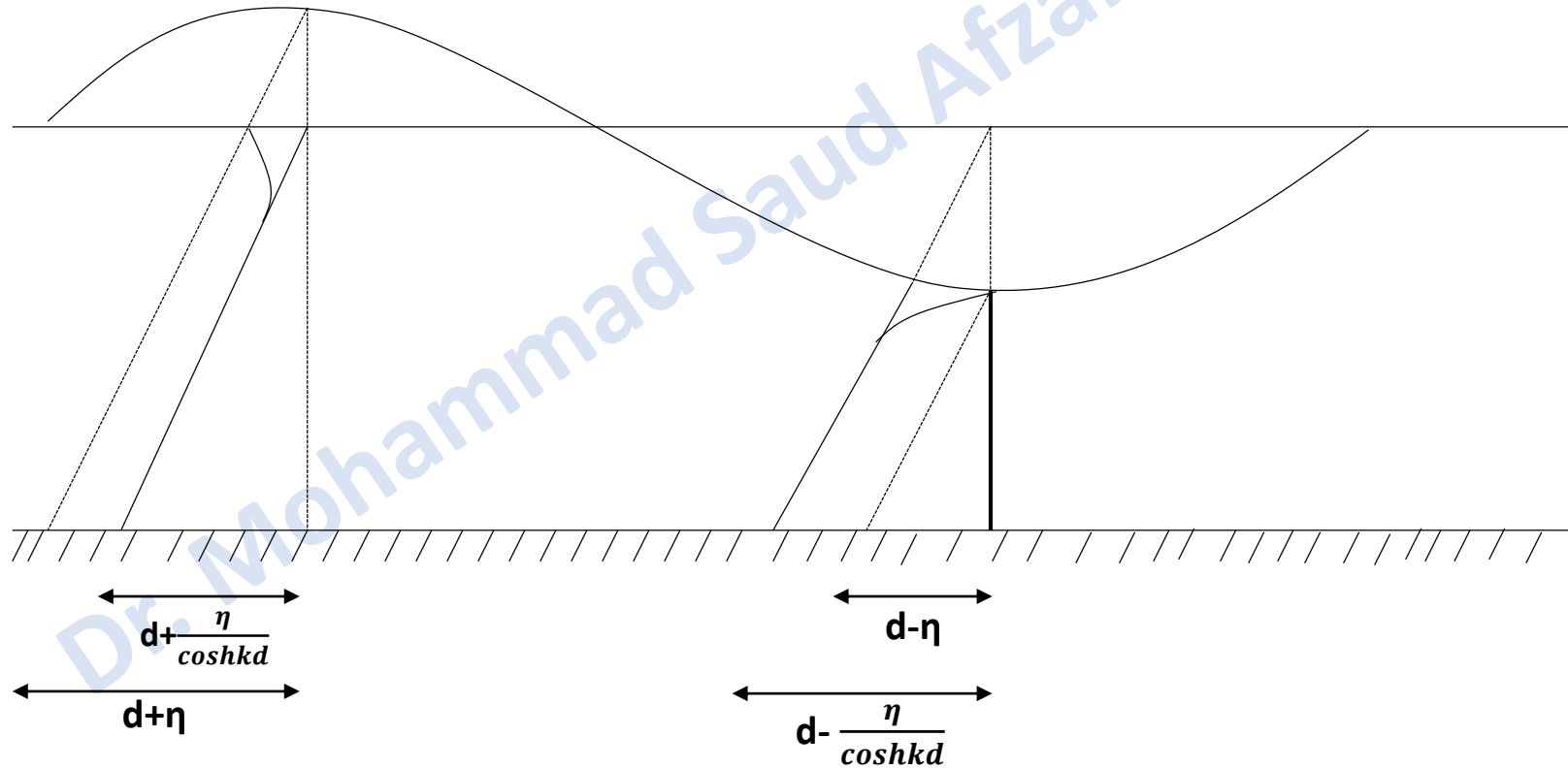


Fig 2.7 Pressure distribution under a progressive wave

GROUP CELERITY:

- When a group of waves or a wave train travels its speed is generally not identical to the speed with individual waves within the group travel. If any two wave trains of the same amplitude, but slightly different wavelengths or periods, progress in the same direction, the resultant surface disturbance can be represented as the sum of the individual disturbances. For waves propagating in deep or transitional waters, the group velocity is determined as follows.

$$\eta_T = \eta_1 + \eta_2 = a \sin(k_1 x - \sigma_1 t) + a \sin(k_2 x - \sigma_2 t) \quad (2.60)$$

$$\eta_T = 2a \cos \left[\left(\frac{k_1 - k_2}{2} \right) x - \left(\frac{\sigma_1 - \sigma_2}{2} \right) t \right] \cdot \sin \left[\left(\frac{k_1 + k_2}{2} \right) x - \left(\frac{\sigma_1 + \sigma_2}{2} \right) t \right]$$

- This is a form of a series of sine waves the amplitude of which varies slowly from 0 to 2a according to the cosine factor.
- The points of zero amplitude(nodes) of the wave envelope η_T are located by finding the zeros of the cosine factor.

i.e

$\eta_T \text{max} = 0$ occurs when

$$\left(\frac{k_1 - k_2}{2}\right) x - \left(\frac{\sigma_1 - \sigma_2}{2}\right) t = (2m+1)\frac{\pi}{2}$$

- In other words the nodes will occur on 'x' axis at distances as follows:

$$X_{node} = \frac{(2m+1)\pi}{k_1 - k_2} + \left(\frac{\sigma_1 - \sigma_2}{k_1 - k_2}\right) t$$

- Since the position of all the nodes in a function of time they are not stationary. At $t=0$, there will be nodes at

$$\frac{\pi}{k_1-k_2}, \frac{3\pi}{k_1-k_2}, \frac{5\pi}{k_1-k_2}, \frac{7\pi}{k_1-k_2} \text{ etc i.e at } m=0,1,2,3....$$

- The distance between the nodes are given by

$$x = \frac{2\pi}{k_1-k_2} = \frac{L_2 L_1}{L_2-L_1} \quad (2.61)$$

- The speed of propagation of the nodes and hence the speed of propagation of the wave group is called the 'Group Velocity' and is given by:

$$\frac{dx_{node}}{dt} = \text{Wave group velocity } C_G = \frac{\sigma_1 - \sigma_2}{k_1 - k_2} = \frac{d\sigma}{dk}$$

But $\sigma = K.C = \frac{2\pi}{L} \frac{L}{T} = \frac{2\pi}{T}$

$$C_G = \frac{d(KC)}{dK} = C + k \frac{dc}{dk} = C + \frac{K.dC}{dL} \left(\frac{1}{\frac{dK}{dL}} \right)$$

Since $k = \frac{2\pi}{L}$

$$C_G = C + \frac{2\pi.dC}{L.dL} \left(\frac{1}{\frac{-2\pi}{L^2}} \right)$$

$$C_G = C - L \frac{dC}{dL}$$

Since $C^2 = \frac{g}{k} \tanh(kd)$

Substituting and on simplification we get

$$\frac{C_G}{C} = n = \frac{1}{2} \left[1 + \frac{2kd}{\sinh 2kd} \right] \quad (2.62)$$

For deep waters $\frac{2kd}{\sinh 2kd}$ is zero

Hence $C_G = \frac{1}{2} C_0$

$$C_G = \frac{1}{2} \frac{L_0}{T} = \frac{1}{2} C_0 \quad (2.63)$$

The Group velocity is one half of the phase velocity in deep waters. Further it should be noted that variables if associated with a suffix '0' refer to deep-water conditions. For example C_0 is deep water celerity.

Table 2.2 Variation of Asymptotic Functions

Function	Asymptotes	
	Shallow waters	Deep Waters
$\sinh kd$	kd	$\frac{e^{kd}}{2}$
$\cosh kd$	1	$\frac{e^{kd}}{2}$
$\tanh kd$	Kd	1

In shallow waters,
since $\sinh 2kd = 2kd$

$$C_G = C = \sqrt{gd} \quad (2.64)$$

Hence in shallow water the group and phase velocities are same and is a function of only depth of water and in deep waters, the C_G is a function of wave length. Because of this, in deep waters, the longer waves (Long L) travel faster and produce the small phase differences resulting in wave groups. These waves are said to be dispersive or propagating in a dispersive medium, i.e., in a medium where their celerity is dependent on wave length.

WAVE ENERGY

- Total Energy = Potential energy + Kinetic Energy
- In order to determine the total energy under progressive waves the potential energy of the wave above $z=-d$ with a wave from present is determined from which the potential energy of the water in the absence of a wave from is subtracted. Refer Fig 2.8 for definitions.
- The potential energy (with respect to $z=-d$) of a small column of water $(d+\eta)$ high, dx long and 1m wide.

$$\begin{aligned} dPE_1 &= \gamma A \bar{x} \\ &= \gamma dx \cdot (d+\eta) \left(\frac{d+\eta}{2} \right) = \gamma \frac{(d+\eta)^2}{2} dx \end{aligned} \quad (2.65)$$

- The average potential energy per unit surface area (sometimes called the average potential energy density) is

$$\overline{PE_1} = \frac{\gamma}{2} \frac{1}{L} \frac{1}{T} \int_t^{t+T} \int_x^{x+L} (d + \eta)^2 dx dt \quad (2.66)$$

Using $\eta = a \sin(kx - \sigma t)$ then Eq. (2.66) becomes

$$\overline{PE_1} = \frac{\gamma}{2LT} \int_t^{t+T} \int_x^{x+L} (d^2 + 2ad \sin(kx - \sigma t) + a^2 \sin^2(kx - \sigma t)) dx dt$$

On simplification

$$\overline{PE_1} = \frac{\gamma d^2}{2} + \frac{\gamma a^2}{4} \quad (2.67)$$

- Which is the average potential energy per unit surface area of all the water above $z=-d$
- The potential energy in the absence of wave would be

$$\overline{PE}_2 = \frac{\gamma}{2LT} \int_t^{t+T} \int_x^{x+L} d^2 dx dt = \frac{\gamma d^2}{2} \quad (2.68)$$

- The average potential energy density, $\overline{PE}$ which is attributable to the presence of the progressive wave on the free surface, is

$$\begin{aligned} \overline{PE} &= \overline{PE}_1 - \overline{PE}_2 = \text{Average Potential Energy} \\ &= \frac{\gamma d^2}{2} + \frac{\gamma a^2}{4} - \frac{\gamma d^2}{2} \end{aligned}$$

$$\overline{PE} = \frac{\gamma a^2}{4} \quad (2.69)$$

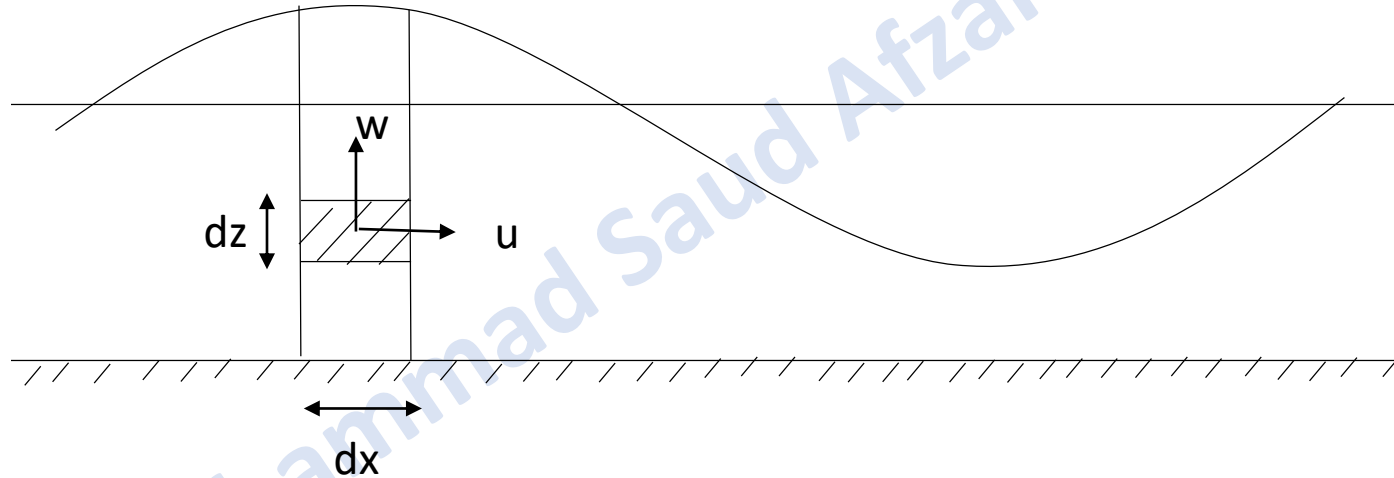


Fig.2.8 Definition sketch for potential and kinetic energy under progressive wave

Kinetic Energy

The kinetic energy, $KE = \frac{1}{2} m v^2$, where 'm' is the mass of the fluid and 'v' is the resultant velocity. For 2D wave flow

$$d(KE) = \frac{1}{2} \left(\sqrt{u^2 + w^2} \right)^2 dM$$

$$= \frac{1}{2} \left(\sqrt{u^2 + w^2} \right)^2 \rho \cdot dz \cdot dx$$

- The average K.E. per unit of surface area is then given by

$$\overline{KE} = \frac{\rho}{2LT} \int_t^{t+T} \int_x^{x+L} \int_{-d}^{\eta} (u^2 + w^2) dz dx dt$$

- With $\eta = a \sin(kx - \sigma t)$

$\overline{KE}$

$$= \frac{\rho}{2LT} \int_t^{t+T} \int_x^{x+L} \int_{-d}^{\eta \cong 0} \frac{(agk)^2}{\sigma^2 \cosh^2 kd} (\cosh^2 k(d+z) \sin^2(kx-\sigma t) + \sinh^2 k(d+z) \cos^2(kx-\sigma t)) dz dx dt$$

Using trigonometrical identities

$$\cosh^2 k(d+z) = \frac{1}{2} [1 + \cosh 2k(d+z)]$$

$$\sinh^2 k(d+z) = \frac{-1}{2} [1 - \cosh 2k(d+z)]$$

$$\cos^2(kx-\sigma t) - \sin^2(kx-\sigma t) = \cos 2(kx-\sigma t)$$

$$\cos^2(kx - \sigma t) + \sin^2(kx - \sigma t) = 1$$

$$\sinh 2kd = 2 \sinh kd \cdot \cosh kd$$

$$\text{And } \sigma^2 = gk \tanh kd$$

$$\text{It can be shown } \overline{KE} = \frac{\gamma a^2}{4} \quad (2.70)$$

$$\text{Total energy } E = \overline{PE} + \overline{KE}$$

$$E = \frac{\gamma a^2}{2} \quad (2.71)$$

The average total energy per unit surface area is the sum of the average potential and kinetic energy densities often called as specific energy density.

WAVE POWER

- Wave energy flux is the rate at which energy is transmitted in the direction of wave propagation across a vertical plane perpendicular to the direction of the wave advance and extending down the entire depth. The average energy flux per unit wave crest width transmitted across a plane perpendicular to wave advance is

$\bar{P}$ = Wave power = Average energy flux per unit wave crest width

$$\bar{P} = \bar{E} n C = \bar{E} C_g \quad (2.72)$$

$$\text{When } n = \frac{1}{2} \left[1 + \frac{2kd}{\sinh 2kd} \right]$$

$$\text{For Deep waters } \frac{2kd}{\sinh 2kd} = 0 \text{ and } C_g = \frac{1}{2} C_0$$

$$n = \frac{1}{2}$$

Or $\overline{P}_0 = \frac{1}{2} \overline{E} C_0 \quad (2.73)$

For shallow waters

$$\overline{P} = \overline{E} C = \overline{E} C_g \quad (\text{since } \sinh 2kd = 2kd)$$

- Assume the wave propagates from deep waters towards the shore. The ocean bottom slope is gradual and there are no undulations and has parallel bottom slope contours. According to the conservation of energy, equating the power in the shallow waters (Eq. 2.72) to that in deep waters (Eq. 2.73) we get

$$\frac{\gamma H^2}{8} \cdot C_g = \frac{\gamma H_0^2}{8} \cdot \frac{C_0}{2}$$

On substituting for C_g and on simplification we obtain

$$\left(\frac{H}{H_0}\right)^2 = \frac{C_0}{C} \frac{1}{\left[1 + \frac{2kd}{\sinh 2kd}\right]}$$

Or

$$\frac{H}{H_0} = \sqrt{\frac{C_0}{C} \cdot \frac{1}{2n}} = k_s \quad (2.74)$$

Where $n = \frac{1}{2} \left[1 + \frac{2kd}{\sinh 2kd} \right]$

- The above equation giving the ratio between wave height at any depth in shallower waters and the deep water height. This relationship obtained without considering the irregular variation in the sea bottom contours is called as *shoaling coefficient*.

The variation of the different properties of a small amplitude waves are shown in Fig.2.9

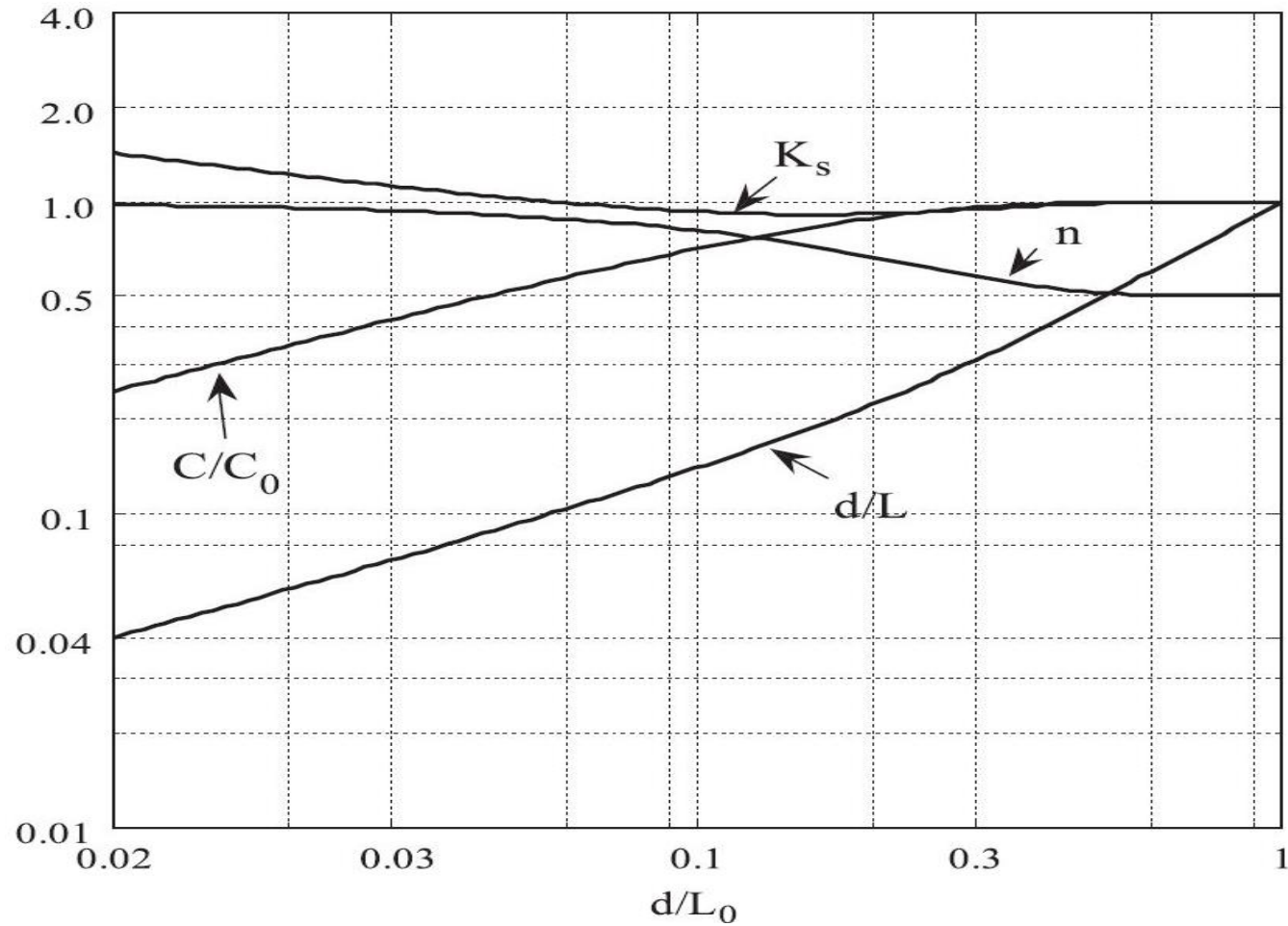


Fig.2.9 Properties of small amplitude waves

MASS TRANSPORT VELOCITY

When waves are in motion, the particles upon completion of each nearly an elliptical or circular motion would have advanced a short distance in the direction of propagation (Fig.2.10). Consequently there is a mass transport in the direction of progress of the wave. The mass transport velocity at any depth z below S.W.L is given as

$$\bar{U}(z) = \left(\frac{\pi H}{L}\right)^2 \frac{C}{2} \frac{\cosh 2k(d+z)}{\sinh^2 kd} \quad (2.75)$$

The mass transport speed is appreciable for high steep waves and is very small for waves of long period.

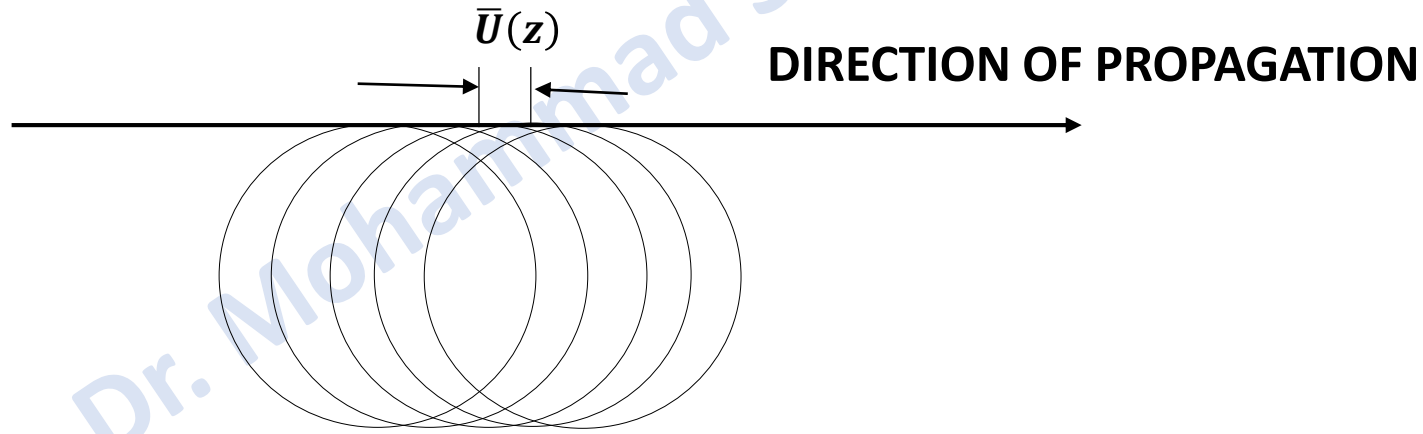


Fig. 2.10